FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Main)-2019

TEST DATE: 10-02-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. D

Sol. $\beta = \frac{I_C}{I_B}$ or $I_B = \frac{I_c}{\beta} = \frac{I_C}{100}$

Also, $V_{CE} = V_{CC} - I_C R_L$

$6 = 10 - I_C \times 1000$

$\Rightarrow I_C = \frac{4}{1000} = 4 \times 10^{-3}\,A$

$\therefore I_B = \frac{I_C}{100} = 4 \times 10^{-5}\,A$

So, $R_B = \frac{V_{CC} - V_{BE}}{I_B} \approx \frac{V_{CC}}{I_B} = \frac{10}{4 \times 10^{-5}} = 2.5 \times 10^5\,\Omega$

2. C

Sol. During elastic collision between two particles, the maximum kinetic energy is transferred from one particle to the other when they have the same mass. Consequently, a neutron loses all its kinetic energy when it collides head on with a proton. For this reason materials which are abundant in hydrogen such as paraffin and water, are good moderates for neutrons.

3. D

Sol. Option, A, B, C are correct and D is the only incorrect option as:

(A) $\lambda = \frac{c}{\nu} = \frac{3 \times 10^8}{10 \times 10^3} = 30$ km

So, minimum length of antenna required $\approx \lambda/4 = 7.5$ km

(B) $P \propto \left(\frac{\ell}{\lambda}\right)^2$

(C) SKY wave communication is used for short wave communication (3MHz – 30 MHz).

(D) Space wave is used to transfer very high frequency (~ 3GHz)

4. D

Sol. Intensity of light after passing through polariser 1 is $I_0/2$.

Intensity of light after passing through polariser 2 is $\frac{I_0}{2}\cos^2\theta$.

Intensity of light after passing through polariser 3 is

$$I = \frac{I_0}{2}\cos^2\theta\cos^2(90-\theta) = \frac{I_0}{2}\sin^2\theta\cos^2\theta$$

So, for $\theta = 0$ or 90°, $I = 0$

for $\theta = 45°$, $I = I_0/8$

for $\theta = 30°$, $I = \frac{I_0}{2}\times\left(\frac{1}{2}\right)^2\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3I_0}{32}$

$\therefore$ d is incorrect.

5. D

Sol. Let a = diameter of the telescope = 50 cm = 0.5 m

d = separation stars = 10^{11} m.

D = distance of stars from earth = 10^5 l_y = $10^5 \times 9.47 \times 10^{15}$ m = 9.47×10^{20} m.

If $\Delta\theta$ is the angle subtended by the line joining the stars with the telescope Then,

$$\Delta\theta = \frac{d}{D} = \frac{10^{11}m}{9.47\times10^{20}m} \approx 10^{-10}\text{rad}$$

Limit of resolution of the telescope

$$\Delta\theta' = 1.22\frac{\lambda}{a} = 1.22\times\frac{600\times10^{-9}}{0.5} = 1.46\times10^{-6}\text{rad}$$

Since, $\Delta\theta << \Delta\theta'$ so the stars cannot be resolved by this telescope or alternately separate images will not be seen.

Note: Information on focal length is not required.

6. D

Sol. Maximum Kinetic energy of β_1^- should be less than maximum kinetic energy of β_2^-. So, option (A) and (B) are incorrect.

Energy associated with γ_1 is 1.088 MeV. So option (C) is incorrect.

From the nuclear energy level diagram of $^{198}_{80}Hg$ it is clear that energy released during emission of γ_3 is (1.088 – 0.412) MeV = 0.676 MeV.

7. C

Sol. Each electron will not only interact with positively charged nucleus (which provides the central force) but also with the other electrons (i.e. we have to take electron-electron interaction). and hence electron's are not subject to central force.

8. C

Sol. Least count = $\frac{0.05}{100}$cm = 0.005 mm

Zero error = $(100 - 5) \times 0.005 = -0.475$ mm

Reading = MSR + CSR $\times$ L.C.

= $4.6 + 52 \times 0.005$

= 4.6 + 0.26 = 4.86 mm
Actual diameter = 4.86 + 0.475 = 5.335 mm

9. C

Sol. $5 \times 10^3 I_1 = 60 \Rightarrow I_1 = \frac{60}{5\times10^3} = 12$ mA

Potential difference across 10 kΩ = 240 − 60 = 180 V
So, $I_3 \times 10 \times 10^3 = 180 \Rightarrow I_3 = 18$ mA
$I_2 = I_3 - I_1 = 6$ mA

10. B

Sol. Total voltage across the circuit $V = I\left[R^2 + (x_L - x_C)^2\right]^{1/2} = IR = 20$ V

When inductor is short circuited

$$I = \frac{20}{\sqrt{R^2 + X_C^2}} = \frac{20}{\sqrt{2}R}$$

$$\therefore\ V_C = IX_C = \frac{20}{\sqrt{2}R} \times R = 10\sqrt{2}\ V$$

11. B

Sol. $\mu = \frac{C_{air}}{C_{median}} = \frac{C}{f\lambda}$ or $\lambda = \frac{C}{f\mu} = \frac{A}{\mu}$ where A is same for the 3 media.

For A : $\lambda_A = \frac{A}{\mu_A}$, for B : $\lambda_B = \frac{A}{1.4}$, for C: $\lambda_C = \frac{A}{1.5}$

So, No. of waves in A = $\frac{t/3}{\lambda_A} = \frac{t\mu_A}{3A}$

No. of waves in B = $\frac{2t/3}{\lambda_B} = \frac{2t}{3A} \times 1.4$

No. of waves in C = $\frac{t}{\lambda_C} = \frac{t \times 1.5}{A}$

Since, the no of waves in C is equal to the sum of no. of waves in A and B, so

$$\frac{1.5t}{A} = \frac{\mu_A t}{3A} + \frac{2.8t}{3A}$$

$$\Rightarrow 1.5 = \frac{\mu_A}{3} + \frac{2.8}{3}$$

$$\mu_A = 1.7$$

12. C

Sol. First diffraction minima is formed at $\frac{\lambda D}{a} = \frac{4500\times10^{10}\times1}{2\times10^{-6}} = 2.25$ mm

(where λ = 4500Å, a = 2 μm, D = 1 m)

Second diffraction minima is formed at $\frac{2\lambda D}{a} = 4.5$ mm

The interference fringe width is $\frac{\lambda D}{d} = \frac{4500\times10^{-10}\times1}{9\times10^{-6}} = 0.5$ mm

So the 9th interference maxima falls on the 2nd diffraction minima. So this will be missing.

13. D

Sol. Flux remains unchanged for sudden change

$\therefore\ L_iI_i = L_fI_f$

$\therefore\ I_f = \dfrac{I_i}{2} = \dfrac{E}{2R}$

$L_i == \dfrac{\mu_0 N^2 A}{L}, L_f == \dfrac{\mu_0\mu_r N^2 A}{L} = 2L_i$

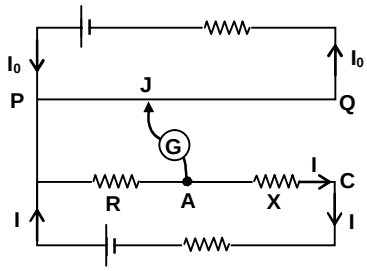
$\therefore$ If just after introducing iron cone is $\dfrac{I_i}{2} = \dfrac{E}{2R}$

(As $I_i = \dfrac{E}{R}$)

$\therefore\ I = \dfrac{E}{2R}e^{-\frac{Rt}{2L}} + \dfrac{E}{R}\left(1 - e^{-\frac{Rt}{2L}}\right) = \dfrac{E}{2R}\left(2 - e^{-\frac{Rt}{2L}}\right)$

14. D

Sol. In potentiometer wire potential difference is directly proportional to the length. Let potential drop per unit length of the potentiometer wire be K. For zero deflection in the galvanometer the current will flow independently in 2 loops.



So, $IR = K \times 20$ …(i)

Also $I(R + x) = K \times 50$ ….(ii)

Subtracting equation (ii) from equation (i)

$IX = K \times 30$ …(iii)

Dividing equation (i) by equation (iii)

$\dfrac{R}{x} = \dfrac{20}{30} = \dfrac{2}{3} \Rightarrow X = \dfrac{3R}{2}$

15. C

Sol. The electric field intensity due to each uniformly charged infinite plane is uniform. The electric field intensity at point A, B, C and D due to plane 1, plane 2 and both planes are given by E_1, E_2 and E as shown in the figure-A. Hence the electric lines of forces are as given in figure –B.

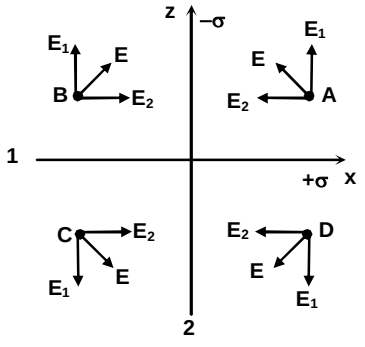


Figure –(A)

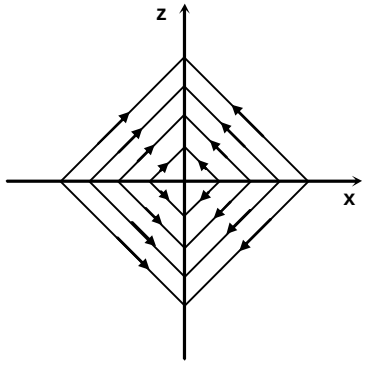


Figure –(B)

Alternate: electric lines of forces originate from positively charged plane and terminate at negatively charged plane. Hence, the correct representation of ELOF is as shown in figure –B.

16. B

Sol. $V_{outside} = \dfrac{KQ}{r}$ …(i)

For E_{in} use gauss law

$$E_{in} = \frac{\rho_0}{\varepsilon_0}\left(\frac{r}{3} - \frac{r^2}{4R_0}\right)$$

Now, $\int_{V_s}^{V_{in}} dV_{in} = \int_{r-R}^{r} \vec{E}_{in} \cdot dr$

$$V_{in} = \frac{\rho_0}{12R_0\varepsilon_0}(r^3 + 2R^3 - 2r^2R)$$

17. A

Sol. The magnitude of velocity can be written as

$v^2 = v_x^2 + v_y^2$

Differentiate

$2v\frac{dv}{dt} = 2v_x\frac{dv_x}{dt} + 2v_y\frac{dv_y}{dt}$

$0 = 2v_xa_x + 2v_ya_y$

So, $v_xa_x + v_ya_y = 0$

18. D

Sol. Apply continuity equation and Bernoulli's equation. At the same horizontal level pressure is less where velocity is more.

19. C

Sol. When block is slightly displaced towards right then

$F_r = -(2kx + 4kx)$

When block is slightly displaced towards left then

$F_r = -(2kx)$

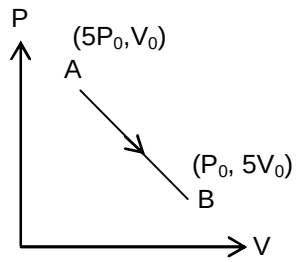
20. C

Sol. System is released then acceleration of 3M and M must be equal. Hence friction force between sphere and plank is zero.

21. A

Sol. Equation of process

$$P = -\frac{P_0}{V_0}V + 6P_0 \quad ...(1)$$



Heat reverses its direction when slope of line is equal to

$$\frac{\gamma P}{V} = \frac{P_0}{V_0} \quad ...(2)$$

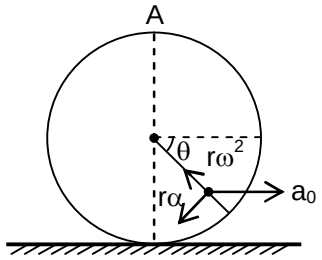
From equation (1) and (2)

$V = \frac{7}{2}V_0, \quad P = \frac{5}{2}P_0$

22. D

Sol. From lami's

$$\frac{a_0}{\sin 90°} = \frac{r\alpha}{\sin(\pi - \theta)} = \frac{r\omega^2}{\sin\left(\frac{\pi}{2} + \theta\right)} \quad ...(i)$$



$a_0 = R\alpha$ ...(ii)

from above equations

$\sin\theta = r/R$

$\theta = 45°$

23. A

Sol. $C = \dfrac{Q^2 \Delta\theta}{m \cdot \gamma}$

$\gamma = \dfrac{Q^2 \cdot \Delta\theta}{m \cdot C} = \dfrac{[ML^2T^{-2}]^2\theta}{[M]\left[\dfrac{ML^2T^{-2}}{M\theta}\right]} = [ML^2T^{-2}\theta^2]$

24. C

Sol. Till 1 sec

$x_A = 2t$

$x_B = t^2$

$x = x_A - x_B = 2t - t^2$

$\dfrac{dx}{dt} = 0$, t = 1 sec, x = 1 m

Now from 1 sec – 5 sec

$x_A = 2 + 4(t - 1)$

$x_B = t^2$

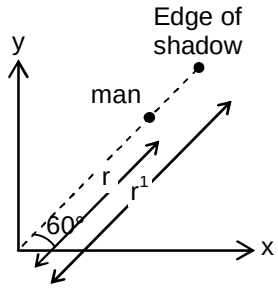
$x = x_A - x_B = 4t - 2 - t^2$

$\dfrac{dx}{dt} = 0$, t = 2 sec

x = 2 m

25. D

Sol. Let at any instant position of man is at r and its edge of shadow is r' then



$\dfrac{2}{r^1 - r} = \dfrac{5}{r^1}$

$3r^1 = 5r$

$\dfrac{3 \cdot dr^1}{dt} = \dfrac{5 \cdot dr}{dt}$

$3.V_s = 5 \times 6$

$V_s = 10$ m/s

$\vec{V}_s = 5\hat{i} + 5\sqrt{3}\hat{j}$

26. A

Sol. $A = \int y dx$

$= \int_0^R \left(x\tan\theta - \dfrac{1}{2}\dfrac{gx^2}{u^2\cos^2\theta} \right) dx$

$= \dfrac{2u^4}{3g^2}\sin^3\theta \cdot \cos\theta$

Now, $\dfrac{dA}{d\theta} = 0$

θ = 60°

27. C

Sol. $v_P = \sqrt{\dfrac{T}{\mu_P}} = 25$ m/s

$$v_Q = \sqrt{\frac{T}{\mu_Q}} = 20 \text{ m/s}$$

$$A_r = \frac{|v_Q - v_P|}{v_Q + v_P} Ai = \frac{2}{9} mm$$

$$A_t = \frac{2v_Q}{v_Q + v_P} Ai = \frac{16}{9} mm$$

28. C

Sol. $E_x = -\frac{\partial V}{\partial x} = -(6yx + 3x^2z) = 0$

$$E_y = -\frac{\partial V}{\partial y} = -(3x^2 - 2z) = -2$$

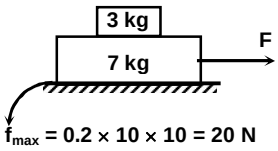
$$E_z = -\frac{\partial V}{\partial z} = -(2y + x^3) = -2$$

$$|\vec{E}| = \sqrt{E_x^2 + E_y^2 + E_z^2} = 2\sqrt{2}$$

29. C

Sol. $a_{common} = \frac{F - 20}{10} > 0$

F > 20 newton



30. C

Sol. $W = \Delta K$

$W_g + W_{spring} = 0$

$$m_1gx - m_2gx - \frac{1}{2}kx^2 = 0$$

$$m_1g - m_2g = \frac{kx}{2}$$

$$x = \frac{2(m_1 - m_2)g}{k} = 2/5 \text{ m.}$$

Chemistry

PART – II

SECTION – A

31. C

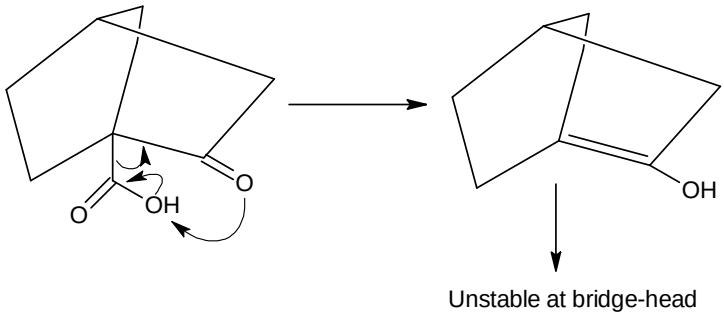
Sol. C_2 (12) = $\sigma 1s^2$ $\sigma^* 1s^2$ $\sigma 2s^2$ $\sigma^* 2s^2$ πp_x^2 πp_y^2

x = 0, y = 2, z = 8

$\therefore \frac{z-x}{y} = \frac{8-0}{2} = 4$

32. C

Sol.



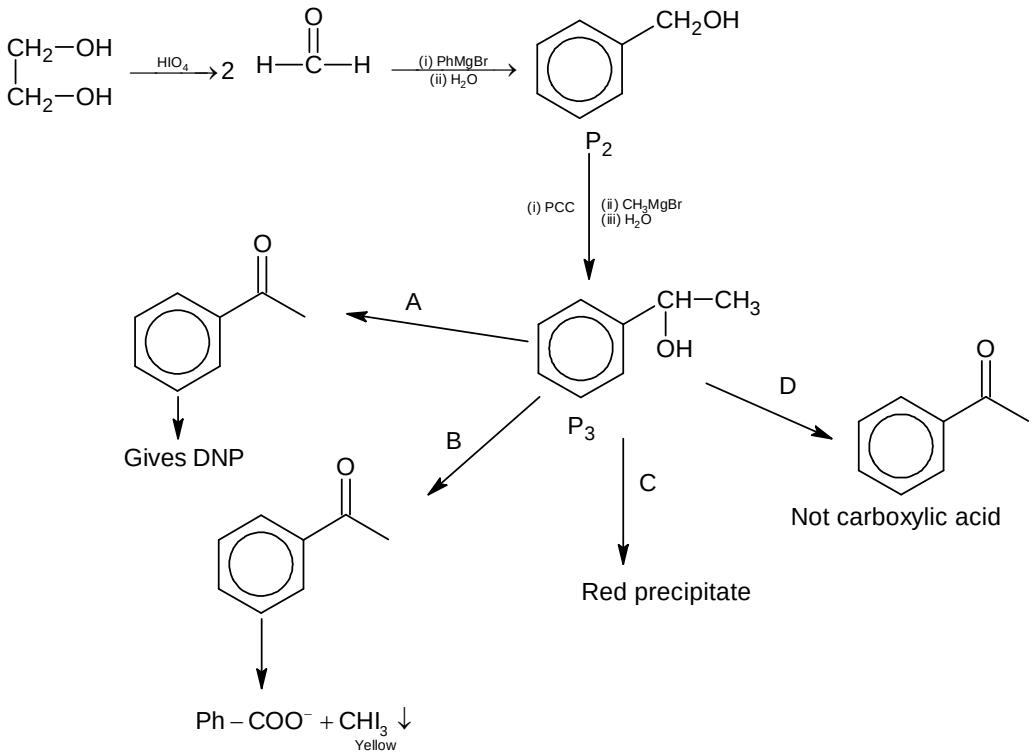
Unstable at bridge-head

33. A

Sol. N^{3-} is the strongest base amongst the following.

34. D

Sol.



35. B

Sol. $\left[Fe(H_2O)_5NO\right]^{+2}$

Fe is in +1 state

n = 3 (unpaired e^-)

$\therefore \mu_s = \sqrt{n(n+2)} = \sqrt{15}$

36. C

Sol.

$$\text{Anode}: \frac{1}{2}H_2 \longrightarrow H^+ + e^-$$

$$\text{Cathode}: Ag^+ + e^- \longrightarrow Ag(s)$$

$$\frac{1}{2}H_2(g) + Ag^+(aq) \longrightarrow H^+(aq) + Ag(s)$$

$$E_{cell} = E^0_{Cell} - \frac{0.059}{1}\log\frac{\left[H^+\right]}{\left[Ag^+\right]}$$

$$Ag_2S \longrightarrow 2Ag^+ + S^{-2}$$

$$Ksp = 4\times10^{-12} = \left[Ag^+\right]^2\left[S^{2-}\right]$$

$$\left[Ag^+\right] = 2\times10^{-4}M$$

$$pH = pKa + \log\left(\frac{salt}{acid}\right)$$

$$= 5 + \log\left(\frac{0.1}{0.1}\right)$$

$$= 5$$

$$E_{cell} = 0.8 - \frac{0.059}{1}\log\left(\frac{10^{-5}}{2\times10^{-4}}\right)$$

= 0.8767 V

37. A

Sol. $$\frac{1}{2}\left(-\frac{dA}{dt}\right) = \frac{1}{3}\left(+\frac{dZ}{dt}\right)$$

$$\frac{dZ}{dt} = \frac{3}{2}\times2\times10^{-4} = 3\times10^{-4}$$

38. D

Sol. $NaBH_4$ reduces aldehydes and ketones not esters.

39. B

Sol. M. P. of diamond ≈ 3930°C.

40. D

Sol.

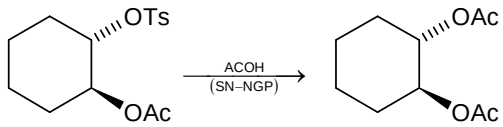
$A \longrightarrow 3B$		$P_\infty = 3P_o \Rightarrow P_o = \frac{P_\infty}{3}$
P_o	–	
$P_o - P$	$3P$	$P_T = P_o + 2P$
–	$3P_o$	$P = \frac{P_T - P_o}{2}$

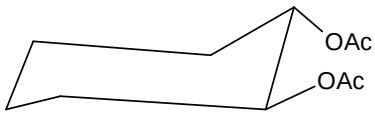
$$K = \frac{1}{t}\ln\left(\frac{P_o}{P_t}\right) \qquad P_t = P_o - P$$

$$= \frac{1}{t}\ln\left\{\frac{P_\infty / 3}{\frac{P_\infty}{3} - \left(\frac{P_T - P_o}{2}\right)}\right\} = \frac{1}{t}\ln\left(\frac{2P_\infty}{3P_\infty - 3P_T}\right)$$

41. A

Sol.





Major product and most stable conformation

42. C

Sol. The elements in middle of the d-block do not form hydrides this is known as hydrogen-gap.

43. B

Sol. I. E_1 = B – 801 in (KJ mol^{-1})
Al – 577
Ga – 579
In – 558
Tl – 589

44. D

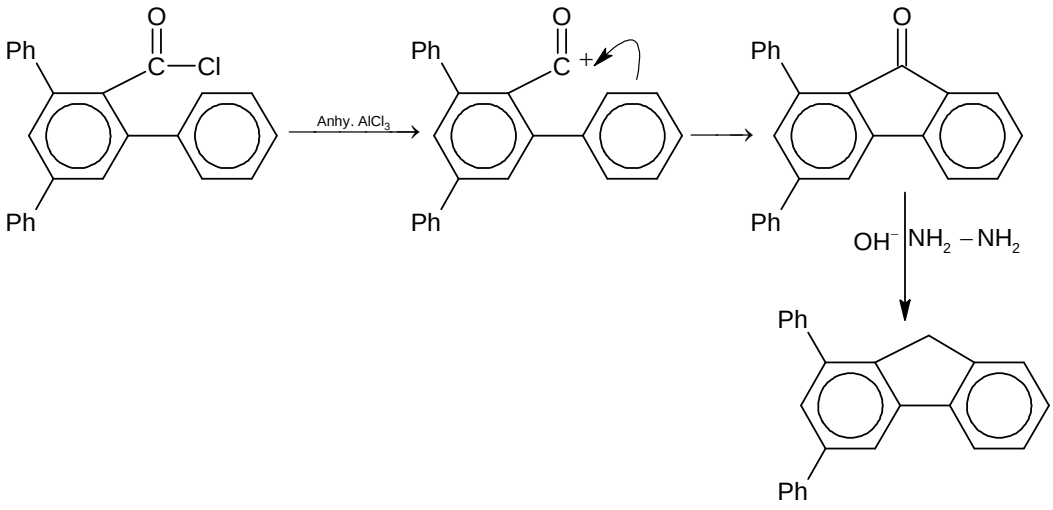
Sol. Trans-$[CoCl_2(en)_2]^+$ contains plane of symmetry.

45. A

Sol. Probability is explained by Quantum theory and not by Bohr Model.

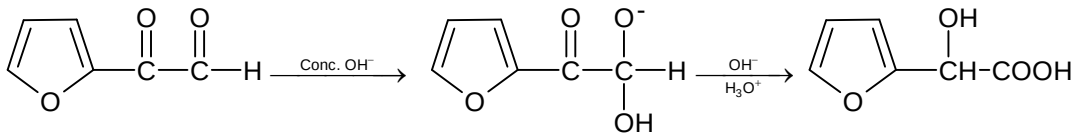
46. C

Sol.



47. C

Sol.



48. B

Sol. $SnCl_2$ – $Na[Sn(OH)_3]$
$CrCl_3 + NaOH$ – $Na_3[Cr(OH)_6]$
$ZnCl_2$ – $Na_2[Zn(OH)_4]$
$BeCl_2$ – $Na_2[Be(OH)_9]$
$AlCl_3$ – $Na[Al(OH)_4]$
All soluble

49. C

Sol. $KClO_3 \longrightarrow KCl + \frac{3}{2}O_2 \uparrow$

$KClO_3$ decomposes (with higher mass) and KCl is formed (lower mass),
$\therefore$ decrease in mass w.r.t. initial stage.

50. A

Sol. $\log\frac{x}{m} = \log k + \frac{1}{n}\log P$

Log k = 0.699

$\therefore$ k = 5

Slope = $\tan\theta = \tan 45° = 1 = \frac{1}{n}$

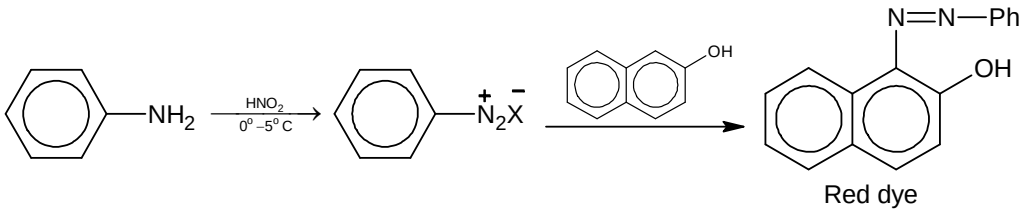
$\therefore \frac{x}{m} = k.p^{\frac{1}{n}}$

$= 5 \times (0.5)^{1/1}$

= 2.5 g/g adsorbent.

51. C

Sol.



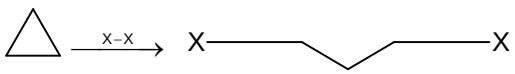
Red dye

52. A

Sol. Copper matte has Cu_2S and FeS along with Cu_2O.

53. D

Sol. (A)

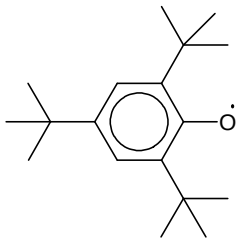


Preffered addition

(B) Reactivity : Chlorination > Bromination.

Reactivity $\propto \frac{1}{\text{selectivity}}$

(C)

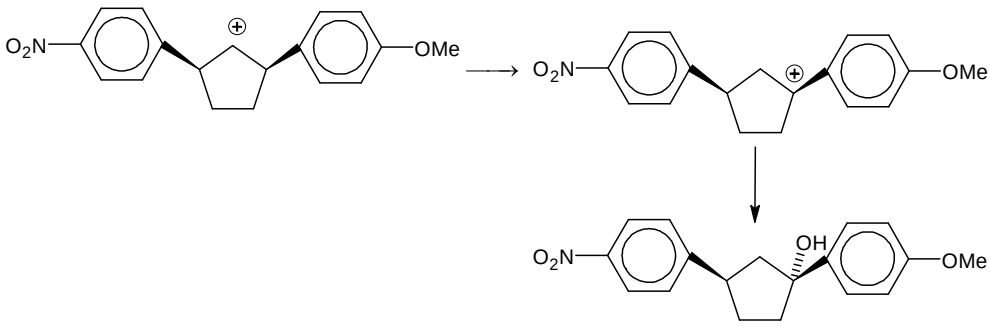


Sterically hindered so cannot give dimerisation reaction.

(D) Boling point = RCl > RF (R = CH_3)
Dipole moment = $CH_3 - F = 1.847$
$CH_3 - Cl = 1.860$

54. D

Sol.



55. A

Sol. $Ca + 2H_2O \longrightarrow H_2 \uparrow + Ca(OH)_2$

$CaH_2 + H_2O \longrightarrow Ca(OH)_2 + 2H_2 \uparrow$

56. B

Sol. $Y_{(CCP)} = 4$ $X_{(Oct.\ void)} = 4$

Atoms along one B. D removed $\therefore\ Y = \frac{15}{4}$, X = 3

Z $\rightarrow$ introduced in available voids = 2 Td. Voids + 1 oct. void = 3

$X_3Y_{\frac{15}{4}}Z_3 \Rightarrow X_4Y_5Z_4$

57. C

Sol. For isothermal process,
$P_1V_1 = P_2V_2$
$32\ P_oV_o = P_oV'$
$V' = 32V_o$
For process R $\rightarrow$ Q
$W = -P\Delta V = -P_o[V_o - V']$
$= -P_o[V_o - 32V_o]$
$W = 31\ P_oV_o$

58. B

Sol. Na + S + C + N $\xrightarrow{\Delta}$ NaSCN (Sodium extract)

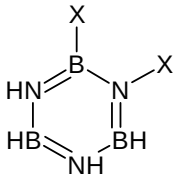
59. D

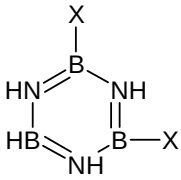
Sol. $NaOH \rightarrow$ High conductance.

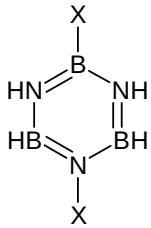
On adding mixture, conductance $\downarrow$, after end point HCl + CH_3COOH mixture will have H^+ from HCl, $\therefore$ conductance $\uparrow$ again.

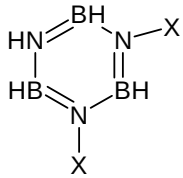
60. D

Sol.









Mathematics

PART – III

SECTION – A

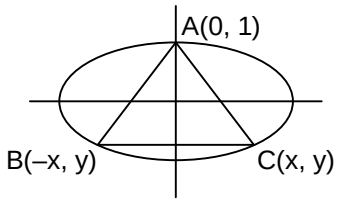
61. A

Sol. Equation of AB

$\Rightarrow \left(y = \sqrt{3}x + 1\right)$

Solve with ellipse

$BC = \frac{16\sqrt{3}}{13}$



62. B

Sol. $\tan^{-1}\left(\frac{4}{r^2 + r + 16}\right) = \tan^{-1}\left(\frac{r+1}{4}\right) - \tan^{-1}\left(\frac{r}{4}\right)$

63. B

Sol. $(x - 1)^2 + (y - 2)^2 + \lambda(x - y + 1) = 0$

Satisfy (1, 0), get λ

64. A

Sol. $x_1^2 + f^2(x_1) - 9 < 0$

65. C

Sol. If (1, 8, 0) occupy odd places $\rightarrow$ 4

If (3, 6, 0) occupy odd places $\rightarrow$ 8

Total : 12

66. B

Sol. $|z_1| = |z_2| = |z_3| = 5$

67. D

Sol. Differentiate both sides, g(x) = tan x

68. A

Sol. $f(x) = \left(\sqrt{x^3 + 1} - 1\right)$ then $f^{-1}(x) = \sqrt[3]{x^2 + 2x}$

69. C

Sol. $\frac{P(e_1e_2e_3) + P(o_1o_2e_1)}{P(e_1e_2e_3) + P(o_1e_1o_2) + P(e_1o_1o_2) + P(o_1o_2e_1)} = \frac{13}{28}$

70. A

Sol. $\sin\alpha = k\cos\frac{\alpha}{2} \Rightarrow \sin\frac{\alpha}{2} = \frac{k}{2}$

$\int_{\alpha}^{\pi}\left(\sin x - k\cos\frac{x}{2}\right)dx = 1 \Rightarrow k = 2 \pm \sqrt{2}$

$-1 \le \sin\frac{\alpha}{2} \le 1,\ k = 2 - \sqrt{2}$

71. C

Sol. $AB = -BA \Rightarrow A^{-1}AB = -A^{-1}BA \Rightarrow B = -A^{-1}BA \Rightarrow BA^{-1} = -A^{-1}B$

$(A^{-1}B)^T = B^T(A^T)^{-1} = -BA^{-1} = A^{-1}B$

72. A

Sol. $f(x) = 1 + x^2\int_0^1 y^2 f(y)\,dy + x\int_0^1 f(y)\,dy$

$f(x) = 1 + ax^2 + bx$, put in the equation and solve, we get $b = f'(0) = \frac{164}{57}$

73. C

Sol. $|a + b + c + d|^2 = \sum|a|^2 + 2\sum \vec{a}\cdot\vec{b} = 4 + 2\vec{d}\cdot\left(\vec{a}+\vec{b}+\vec{c}\right)$

Let $d = \lambda\vec{a} + \mu\vec{b} + \nu\vec{c}$

$\lambda = \mu = \nu = \cos\alpha \Rightarrow \cos\alpha = \frac{1}{\sqrt{3}}$ as $\alpha \in \left(0, \frac{\pi}{2}\right) \Rightarrow |a + b + c + d|^2 = 4 + 2\sqrt{3}$

74. D

Sol. g(x) is increasing $\int_1^6 g'(x)\,dx \le \int_1^6 \frac{2}{x}dx \Rightarrow g(6) - g(1) \le 2\ell n6$

75. C

Sol. $\sin\left(\frac{\pi}{12}\right) = \frac{\sqrt{6}-\sqrt{2}}{4}$, $\cos\frac{\pi}{12} = \frac{\sqrt{6}+\sqrt{2}}{4}$

$\sin\left(x + \frac{\pi}{12}\right) = \sin 2x$

$x = \frac{\pi}{12}, \frac{11\pi}{36}$

76. D

Sol. $T = \int_0^{\ell n2} \frac{\left(2e^{2x} + e^x - e^{-x}\right)}{\left(e^{2x} + e^x - 1 + e^{-x}\right)}dx$

$T \Rightarrow \ell n\frac{11}{4} \Rightarrow T = \frac{11}{4}$

77. A

Sol. $F(x) = F(x + 1)$

$\int_0^{1500} F(x)\,dx = 1500\left(\int_0^{1/2} F(x)\,dx + \int_{1/2}^1 F(x)\,dx\right)$

Now put $x = y + \frac{1}{2}$ in 2nd integral

78. B

Sol. $\lim_{n\to\infty}\left(\frac{1}{2n+1} + \frac{1}{2n+2} + \frac{1}{2n+3} + ... + \frac{1}{4n}\right) - \lim_{n\to\infty}\left(\frac{1}{2n+2} + \frac{1}{2n+4} + ... + \frac{1}{4n}\right)$

$\Rightarrow \int_0^2 \frac{dx}{2+x} - \int_0^1 \frac{dx}{2+2x}dx = \frac{1}{2}\ell n2$

79. B

Sol. $P \equiv (6 + 8i)$, apply rotation, $R = (3 + 7i)$

80. D

Sol. $a + b = 0 \Rightarrow a = -4, b = 4$

Area $= 2\int_0^{1/4} \sqrt{x}\,dx = \frac{1}{6}$

81. B

Sol. Plane is $2x + 3y + 6z = 35$

82. C

Sol. All alike + (2 alike, 2 alike, 2 alike) + (2 alike + 4 alike) + (3 alike + 3 alike) = 1405

83. B

Sol. Variance is independent of shifting of origin

84. A

Sol. $\sim(p \to q) \to (q \to p) \equiv T$

and $\sim\{(p \to q) \wedge (\sim q \wedge p)\} \equiv \sim\{(p \to q) \wedge (\sim(p \to q))\} \equiv T$

85. A

Sol. Let $I = \int_0^1 \ell nx \cdot \ell n(1-x)\,dx = -\int_0^1 \ell nx \sum_{n=1}^{\infty} \frac{x^n}{n} dx$

$\Rightarrow -\sum_{n=1}^{\infty} \frac{1}{n} \int_0^1 x^n \ell nx\,dx$

Use integration by parts, then

$I = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right) - \sum_{n=1}^{\infty} \frac{1}{(n+1)^2}$

$\Rightarrow I = 2 - \frac{\pi^2}{6}$, $p + q + r = 10$

86. D

Sol. $AP = x \cot 60^\circ$

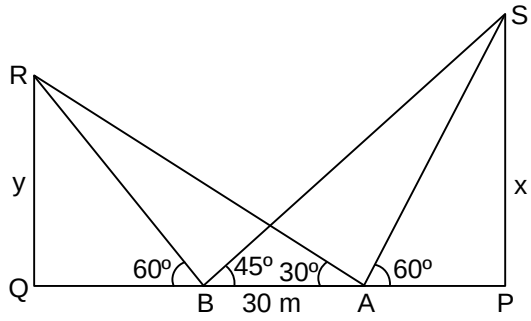
$BP = x \cot 45^\circ$

$\Rightarrow PB - AP = AB$

$30\sqrt{3} = \left(\sqrt{3} - 1\right)x$

Similarly, $y = 15\sqrt{3}$

$\Rightarrow PQ = \left(60 + 15\sqrt{3}\right)m$



87. A

Sol. $AC \cdot BC = SC^2 = \left(\frac{4}{\sqrt{2}}\right)^2$

88. B

Sol. AOCB is a rhombus $h^2 + k^2 = 4r^2$

89. B

Sol. Given expression is $\prod_{r=1}^{n}\left(\frac{8r^2-12r+5}{8r^2+4r+1}\right)=\frac{1}{8n^2+4n+1}$

$k_1 = 8, k_2 = 4, k_3 = 1$

90. B

Sol. $\lim_{n\to\infty} n\tan\left(4\pi n!\left(1+\frac{1}{1!}+\frac{1}{2!}+.....+\frac{1}{n!}+\frac{1}{(n+1)!}+.....\infty\right)\right)$

$$= \lim_{n\to\infty} \frac{n\tan\left(4\pi\left(\frac{n!}{(n+1)!}+\frac{n!}{(n+2)!}+.....\infty\right)\right)}{4\pi\left(\frac{n!}{(n+1)!}+.....\infty\right)} \times 4\pi\left(\frac{n!}{(n+1)!}+.....\infty\right) = 4\pi = L$$